



Birla Institute of Technology & Science, Pilani

Hyderabad Campus

SUPPLEMENTARY ANSWER BOOK

ID No. _____

Name _____

3B
PE Tutorials
Final Solutions.

Course No. _____

Course Title _____

Date _____

Start Writing from here :-

Tutorial 1

Q1) Forward anode current $I_D = 0.675 \text{ mA}$
Thermal voltage $V_T = 0.0263 \text{ V}$

Q2) Rate of change of current $di/dt = 13.158 \text{ A/}\mu\text{s}$
as $t_{\text{or}} = 3.8 \mu\text{s}$ it is hard recovery type

Q3) $i(\omega t) = \frac{I_m}{\pi} + \frac{I_m}{2} \sin \omega t - \frac{2I_m}{3\pi} \cos 2\omega t - \frac{2I_m}{15\pi} \cos 4\omega t - \dots$

Q4) $V_o(\omega t) = \frac{2V_m}{\pi} - \frac{4V_m}{3\pi} \cos \omega t - \frac{4V_m}{15\pi} \cos 4\omega t - \dots$

Q5) $P_{\text{orms}} = 500 \text{ W}$

(a) $I_m = 6.151 \text{ A}$

(b) $\eta_{\text{rect}} = 40.54\%$

(c) $V_{\text{ACrms}} = 125.42 \text{ V}$

(d) $FF = 1.59$

(e) $RF = 1.21$ or 121%

(f) Input $Pf = 0.707$ lagging

- Q6) (a) $I_m = 0.2034 A$
 (b) $I_{DC} = 0.065 A$
 (c) $I_{RMS} = 0.1017 A$
 (d) $E_{DC} = 48.75 V$
 (e) $P_{AC} = 7.912 W$
 (f) $R_F = 1.203$
 (g) $\%R = 1.574 \%$

Tutorial 2

Q1) (a) $V_{OAVG} = 207.073 V$

$I_{OAVG} = 20.707 A$

(b) $I_{OAVG} = 10.3535 A$

$I_{DRMS} = 14.642 A$

(c) $I_{ORMS} = 20.707 A$

$I_{SRMS} = 20.707 A$

(d) $i_s(\omega t) = \frac{4 I_a}{\pi} \left[\frac{\sin \omega t}{1} + \frac{\sin 3\omega t}{3} + \frac{\sin 5\omega t}{5} + \dots \right]$

$\eta_{HD} = 48.43 \%$

Q2) $I_m = 62.832 A$

$P_{IV} = -628.319 W$

$\eta_{UF} = 81.08 \%$

Q3) V_u at 60Hz = $9.596 \times 10^3 V$ $\therefore$ % reduction in voltage = 0.0531%

V_u at 20Hz = 10.816V $\therefore$ % reduction in voltage = 9.54%

Q4) (a) $I_m = 31.13 A$, $V_m = 311.126 V$

$V_{RMS} = 220 V$

$I_{RMS} = 22 A$

$P_{IV} = \underline{\underline{-311.126 V}}$

(b) $V_{OAVG} = 196.468 V$

(c) condition (a) $I_m = 37.335 A$, $N_m = 373.35 V$

$V_{RMS} = 264 V$

$I_{RMS} = 26.4 A$

$P_{IV} = -373.35 V$

condition (b) $V_{OAVG} = 235.76 V$

Table 3

Q1) (a) $V_{OAVG} = 140.348 V$

(b) $I_{OAVG} = 23.39 A$

$I_{PAVG} = 7.797 A$

(c) $P_{IV} = -293.938 V$

(d) $FF = 1.0165$

(e) $RF = 18\%$ or 0.1826

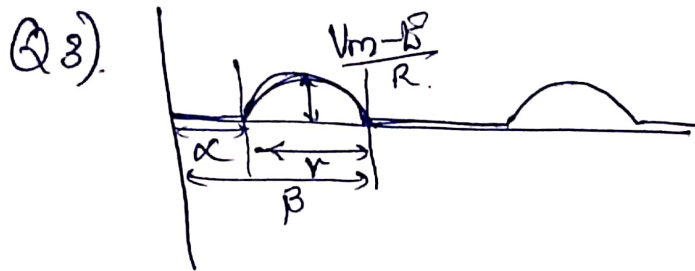
(f) $CF = 1.1896$

$$Q2) V_{OAVG} = 280.693V$$

$$PIV = -293.938V$$

$$I_{SRMS} = 48.989A$$

$$R = 4.678\Omega$$



$$(a) \gamma = 163.74^\circ$$

$$(b) R = 4.256\Omega$$

$$(c) P_R = 287.061W$$

$$(d) h_o = 1 \text{ hr } 40 \text{ min } 12 \text{ sec}$$

$$(e) \eta_{rect} = 17.3\%$$

$$(d) PIV = -96.853V$$

$$Q4) (a) (i) I_{OAVG} = 6.36A$$

$$(ii) P_R = 463.91W$$

$$(b) (i) I_{DAVG} = 3.183A$$

$$(ii) I_{DRMS} = 4.501A$$

Tutorial 4

Q1) $V_{OAVG} = 118.795V$, $V_{ORMS} = 197.098V$

$I_{OAVG} = 2.375A$

$I_{ORMS} = 3.942A$

Q2) (a) $V_{OAVG} = 51.968V$

$I_{OAVG} = 3.45A$

$V_{ORMS} = 114.99V$

$I_{ORMS} = 7.67A$

(b) $\eta_{\%} = 20.25\%$

(c) $ff = 2.22$

$RF = 1.983$

(d) VA rating of transformer = $1764.1 VA$ or $1.764 KVA$

(e) $TUF = 0.10124$

(f) $PW = -328.29V$

Q3) $\text{Lag angle } \alpha = 135^\circ$

Q4) $V_{OAVG} = \frac{V_m}{\pi} (1 + \cos \alpha)$

$V_{OAVG} = 115.27V$, $I_{OAVG} = 3.842A$

$V_{ORMS} = V_m \left[\left(\frac{\pi - \alpha}{2\pi} \right) + \left(\frac{\sin 2\alpha}{4\pi} \right) \right]^{1/2} = 143.024V$

$I_{ORMS} = 4.76A$

Q5) $\alpha = 60.5^\circ$

$$V_{OAVG} = 154.497V$$

$$I_{OAVG} = 2.92A$$

$$I_{ORMS} = 3.889A$$

$$V_{ORMS} = 205.718V$$

Exercice 5

Q1) $V_{OAVG} = \frac{V_m}{2\pi} [1 + \cos \alpha]$

$$V_{ORMS} = V_m \left[\left(\frac{\pi - \alpha}{4\pi} \right) + \left(\frac{\sin 2\alpha}{8\pi} \right) \right]^{1/2}, \quad V_m = 230\sqrt{2}$$

For each value of

$$I_{OAVG} = V_{OAVG} / R$$

$$I_{ORMS} = V_{ORMS} / R$$

$$\eta = \frac{V_{OAVG} \times I_{OAVG}}{V_{ORMS} \times I_{ORMS}} \times 100\%$$

$$FP = \frac{V_{ORMS}}{V_{OAVG}}$$

PIV for each waveform.

$$RP = \sqrt{FP^2 - 1}$$

For each value of α determine all the above.

Q2)

α	$V_{OAVG} (V)$
0°	105.03
45°	73.394
90°	-3

$$\alpha = 88.41^\circ$$

$$PIV = -168.205V$$

Q3) $N = 13418 \text{ rpm}$

Q4)(a) $t_c = 8.33 \text{ msec}$

$$V_{OAVG} = 84.477V$$

$$I_{OAVG} = 16.898A$$

(b) $t_c = 9.431 \text{ msec}$

$$V_{OAVG} = 142.544V$$

$$I_{OAVG} = 6.51A$$

Q5)(a) $\alpha = 53.214^\circ$

(b) $\alpha = 124.069^\circ$

In (a) power is utilized by load from source (P is $+ve$)

(b) Power is provided from a load to a source (P is $-ve$)

(c) $\cos \phi$ case (a) is 0.539 lagging

(b) is -0.504 lagging

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